

Vinh Ly

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Summary

Software Engineer with **18 years** of experience designing and shipping distributed systems at Google, YouTube, Whatnot, and Microsoft. Built search infrastructure serving billions of queries, video streaming pipelines at scale, and real-time marketplace platforms handling millions in transactions. Led teams of 6-10 engineers across three organizations. Focusing on high-throughput backend systems and platform reliability. Owns production infrastructure end to end, including on call coverage, deployment automation, and monitoring across distributed services. Comfortable partnering with platform and application teams on reliability work.

Core Skills

Languages and Frameworks: TypeScript, JavaScript, HTML, CSS

Distributed Systems: consensus protocols, sharding, caching layers, rate limiting

Backend Engineering: Python, Java, C++, Go, gRPC, Protocol Buffers, REST APIs

Infrastructure: Kubernetes, Docker, Terraform, GCP, AWS, Linux

Data Engineering: Kafka, Bigtable, Spanner, SQL, data pipelines

Work Experience

Staff Software Engineer | YouTube | 02/2025 - Present

- Designed video ingestion and transcoding pipelines handling millions of concurrent uploads with sub-second latency targets.
- Led cross-functional initiative to reduce video start time by 40% through pipeline optimization and prefetching.
- Implemented global CDN orchestration layer reducing origin load by 35% through intelligent edge caching.
- Built adaptive bitrate streaming infrastructure supporting 4K playback across mobile and TV platforms.
- Established reliability standards for video infrastructure achieving 99.99% uptime across all streaming services.

Staff Software Engineer | Whatnot | 05/2024 - 02/2025

- Built auction and bidding system processing \$50M+ in daily transaction volume with sub-100ms update latency.
- Led migration of monolith services to microservices architecture improving deployment frequency from weekly to daily.
- Designed payment processing pipeline achieving PCI DSS compliance and reducing transaction failures by 60%.
- Established on-call rotation and incident response procedures for a 24/7 live commerce platform.
- Architected real-time marketplace platform supporting 500K+ concurrent users during live commerce events.

Staff Software Engineer/Manager | Google | 03/2015 - 04/2024

- Led search serving infrastructure serving 100B+ daily queries across 190 countries with p99 latency under 200ms.
- Managed team of 8 engineers building ranking and relevance systems for core search product.
- Built internal experimentation platform enabling 1000+ concurrent A/B tests across search product lines.
- Reduced infrastructure costs by \$15M annually through capacity optimization and workload migration.

- Designed feature pipeline processing 10TB daily for machine learning ranking models.

Software Engineer | Google | 02/2011 - 02/2015

- Implemented near-real-time search index updates reducing document freshness lag from hours to minutes.
- Developed distributed indexing system for web crawl processing handling 20B documents daily.
- Built MapReduce workflows for large-scale data analysis supporting search quality evaluation.
- Designed spam detection infrastructure filtering 15% of crawl volume before indexing.

Software Design Engineer II | Microsoft | 10/2007 - 12/2011

- Built diagnostic tooling for kernel-level performance analysis reducing issue resolution time by 50%.
- Developed COM and .NET interop libraries exposing low-level OS capabilities to managed code.
- Designed and implemented Windows API frameworks used by millions of third-party applications.
- Authored technical documentation and API specifications for external developer ecosystem.

Software Design Engineer Intern | Microsoft | 06/2006 - 08/2006

- Developed internal testing framework for Windows Shell components improving test coverage by 30%.
- Automated regression test suite reducing manual QA effort by 20 hours weekly.

Education

B.S. Computer Science

Virginia Tech | 2004 - 2007